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Dear Professor Fefferman,

I write only to let you know of an article I have posted to arXiv, *Constructive Existence and Qualitative Uniqueness of Turbulent Navier–Stokes Solutions* (arXiv:XXXX.XXXXX), since it engages directly with your formulation of the problem and quotes you in it.

The article presents what an automated adaptive finite element method delivers when directed at alternative (A): for given data at high Reynolds number, turbulent solutions exhibited by construction, with the error in mean-value outputs such as drag and lift bounded by duality rather than estimated. I am explicit that this is not a discharge of (A) as you state it — what is obtained is constructive existence and qualitative uniqueness, not smoothness, the solutions in question having gradients that grow like $\nu^{-1/2}$ while their velocities stay bounded.

I quote your sentence that the four alternatives are offered “to give reasonable leeway to solvers while retaining the heart of the problem”, and take the Institute’s own description to locate that heart in turbulence. My claim is simply that I have used the leeway to reach turbulence. The material itself is not new: it was published in 2007, in a chapter of a book written with Johan Hoffman.

Sending this is courtesy rather than a submission, and nothing in it needs a reply.

Yours sincerely,

Claes Johnson